Tolu Mohammed

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Skills

Controls & Automation: HMI/SCADA interfacing, control logic development, **Simulink**, **MATLAB**, **CANalyzer**, **P&ID** and **PFD**, **Aspen Plus**.
Commissioning & Project Execution: Commissioning and **SAT** support, Management of Change (**MOC**), OEM documentation review, **MS Project** (Gantt), cross-functional coordination, operator and maintenance training.
Reliability & Continuous Improvement: Root Cause Analysis (**RCA**), failure mode analysis, **MTBF**, cycle time analysis, time study, **SAP EAM**, **DMAIC**, **IL6S**, **5S**, Kaizen.
Data & Compliance: Python data analysis, **VBA**, **AutoCAD**; **GMP**, **SQF** audit, **WHMIS**, SOP development.

Experience

Incoming Automation Engineering Intern — Starting January 2027
Suncor Energy — *Sherwood Park, AB*

Supply Chain Intern (Process & Operations) — January 2026 – April 2026
Mondelez International — *Scarborough, ON*

- Supported new palletizer installation and commissioning, reviewing **OEM** manuals, equipment specifications, and **MOC** documentation with the project team and verifying the install met requirements ahead of **SAT** testing.
- Diagnosed palletizer faults by interfacing with the **HMI** to identify system-level failure modes, escalating findings to the controls team, who rewrote the affected **SCADA** logic.
- Reduced palletizer downtime by conducting **RCA**, working-principle, and cycle time analysis on conveyor jams at the merger across **3** production lines, isolating delay sources and increasing **MTBF** by **12%** toward a **100**-minute target.
- Delivered cross-functional palletizer training to maintenance and warehouse departments in preparation for **SAT** testing.
- Developed an automated Excel planning tool using **VBA**, used weekly to plan inbound materials and outbound production across **6** production lines, cutting manual data-entry errors.
- Led warehouse **5S** and Kaizen initiatives with contractors, improving audit readiness toward Mondelez **CBN**; standardized operations via **SOPs** and **OPLs** in compliance with **GMP**.

Manufacturing Engineering Intern — May 2025 – August 2025
Hood Packaging Corporation — *Burlington, ON*

- Logged **1,200** maintenance parts for **8** production lines in **SAP EAM**, improving stock valuation and tracking; ran time studies on repair turnaround to verify a **40%** improvement over **2** months.
- Developed a plant-wide chemical storage standard across **9** storage areas, categorizing **70+** chemicals with color-coded bins, **WHMIS** signage, and **PPE** guides, achieving food-grade segregation and passing the **SQF** audit.

Process & Continuous Improvement Intern — September 2024 – December 2024
Hood Packaging Corporation — *Burlington, ON*

- Developed a plant-wide order naming convention and **SOP**, streamlining specifications and reducing order delays by **17%**.
- Built an automated **VBA** "Order Name Generator" for **2,150** job orders, cutting order redundancy by **68%**.
- Updated **3** plant layout drawings in **AutoCAD** to reflect **5** new equipment installations and removals.

Propulsion Controls and Modelling Team Member — February 2024 – March 2024
University of Waterloo EcoCAR Design Team (UWAFT) — *Waterloo, ON*

- Researched and implemented boost mode control logic into the vehicle's control system to enhance propulsion performance.
- Investigated **CANalyzer** for Diagnostic Trouble Code (**DTC**) creation, developing a comprehensive guide for the controls team.

Projects

Hydrogen Sailors | *Excel, MS Project*

- Designed and built a hydrogen fuel cell boat, analyzing fuel-cell efficiency across **4** prototype reports.
- Led the team using **MS Project** Gantt charts to optimize task assignments and track project progress.

Coin Cell Lab | *Battery Testing*

- Assembled and tested coin cells across multiple electrolytes, characterizing steady-state voltage and identifying **KOH** as the top performer.
- Ran **12 V** lead-acid discharge trials comparing low- and high-rate charging, demonstrating improved voltage stability under gradual charge.

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Chemical Engineering — *Graduation 2028*

Relevant Coursework: Process Control (in progress), Process Data Analysis, Chemical Reaction Engineering, Heat & Mass Transfer, Thermodynamics, Electrochemical Engineering (in progress).

Leadership Experience

NSBE UWaterloo President: Leading a **100+** member chapter's strategic direction, executive team, and partnerships, raising over **$21,000** in corporate sponsorships while growing attendance and representing the chapter at regional and national forums.
NSBE ECZ Vice-Chair: Supporting strategy and development across Eastern Canada, coordinating leadership initiatives and conference planning with zonal chapter presidents.